Approximate Delay-and-Sum Beamforming for Diverging-Wave Imaging of Pipes

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***Abstract*—Rapid nondestructive testing of liquid-filled steel pipes can be accomplished using diverging-wave (DW) ultrasound imaging, where an immersed phased-array probe emits several DW pulses at different steering angles and records the resulting backscattered signals. This work presents an approximate delay-and-sum beamforming method for simplified multimode image reconstruction of inspected pipe sections. The targeted application context is characterization of surface-breaking cracks in pipe walls, and we show that our approach yields sufficiently accurate images for subsequent defect localization and sizing.**

***Index Terms*—ultrasonic nondestructive testing, ultrafast imaging, receive beamforming.**

I. Introduction

Ultrafast ultrasound imaging enables rapid data acquisition with a small number of transmit events and has proven to be highly effective for biomedical diagnostics [1]. This approach has also been successfully adapted to industrial nondestructive testing applications (e.g., see [2]–[6]). Here, we focus on the multimode image reconstruction task that arises in the context of rapid in-line inspection of liquid-filled large-diameter pipes, which utilizes diverging waves (DW) emitted by an immersed phased-array probe [7]. This work contributes an approximate time-domain receive beamforming method to simplify image reconstruction of the inspected pipe sections, while preserving sufficient accuracy for subsequent localization and sizing of defects, such as backwall cracks.

The setup for DW imaging data acquisition is illustrated in Fig. 1, where the pipe frontwall (inner) and backwall (outer) radii are denoted by R_{fw} and R_{bw}, respectively. The origin of our (z, x) coordinate system is the center (point O) of an immersed phased-array probe. Ideally, the probe center should be at O′, positioned at some nominal standoff distance of d_{fw} and vertically aligned with the pipe center C. The difference between O′ and O (the actual probe center), caused by potential probe misalignment and characterized by the axial and lateral offsets z_{err} and x_{err}, must be taken into account during image reconstruction [8].

In this work, we use delay-and-sum (DAS) beamforming [9] to reconstruct multimode images of insonified pipe sections. Let $\mathcal{A}_\theta(t, x_e)$ represent a 2D frame of channel data acquired over time t by the array elements at locations x_e, after a DW emission steered at some angle θ. In its simplest form, a DAS beamformer reconstructs an image $\mathcal{I}_\theta(z, x)$ from $\mathcal{A}_\theta(t, x_e)$ as follows:

$$\mathcal{I}_\theta(z,x) = \sum_{x_e} w_\theta(z,x,x_e)\mathcal{A}_\theta(t=\tau_\theta(z,x,x_e),x_e), \quad (1)$$

where $w_\theta(z, x, x_e)$ refers to the aperture apodization weights, and $\tau_\theta(z, x, x_e)$ represents applied delays. The latter consists of the transmit-path (TX) and receive-path (RX) components that depend on the imaging mode:

$$\tau_\theta(z,x,x_e) = \tau_{\text{TX}}^{\text{mode}}(z,x,\theta) + \tau_{\text{RX}}^{\text{mode}}(z,x,x_e). \quad (2)$$

In this work, we focus on the L-L, T-T, and TT-T modes, or views, where L and T refer to the longitudinal and transverse wave propagation in steel, respectively. The hyphen conveys an interaction with a target, which separates transmit and receive paths. Thus, "L-" and "T-" indicate the use of *direct* transmit paths to the target, while "TT-" indicates the use of *half-skip* transmit paths. In the latter case, the target is reached after an incoming wave reflects from the backwall. Similarly, "-L" and "-T" indicate the use of direct receive paths from the target to the array elements.[1] To summarize, we compute the total delay as follows.

$$\begin{aligned}
\text{L-L mode:}\quad & \tau_\theta(z,x,x_e) = \tau_{\text{TX}}^{\text{L-}}(z,x,\theta) + \tau_{\text{RX}}^{\text{-L}}(z,x,x_e),\\
\text{T-T mode:}\quad & \tau_\theta(z,x,x_e) = \tau_{\text{TX}}^{\text{T-}}(z,x,\theta) + \tau_{\text{RX}}^{\text{-T}}(z,x,x_e),\\
\text{TT-T mode:}\quad & \tau_\theta(z,x,x_e) = \tau_{\text{TX}}^{\text{TT-}}(z,x,\theta) + \tau_{\text{RX}}^{\text{-T}}(z,x,x_e).
\end{aligned}$$

Our proposed approximations simplify calculations of these delays, and we show that our approach produces sufficiently accurate L-L, T-T, and TT-T views of imaged large-diameter pipe sections.

II. Proposed Method

The approximate DAS beamforming method proposed here involves the following steps:

1. Estimate c_l, the speed-of-sound (SoS) in liquid.
2. Estimate z_{err} and x_{err}, the probe positioning errors.
3. Estimate $c_{s\text{L}}$ and $c_{s\text{T}}$, the longitudinal and transverse SoS in steel.

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[1] There are many other modes that can be used during image reconstruction (e.g., LL-LL, T-TT, TT-L, etc.). We focus on the L-L, T-T, and TT-T modes as they are particularly well-suited for imaging backwall cracks.

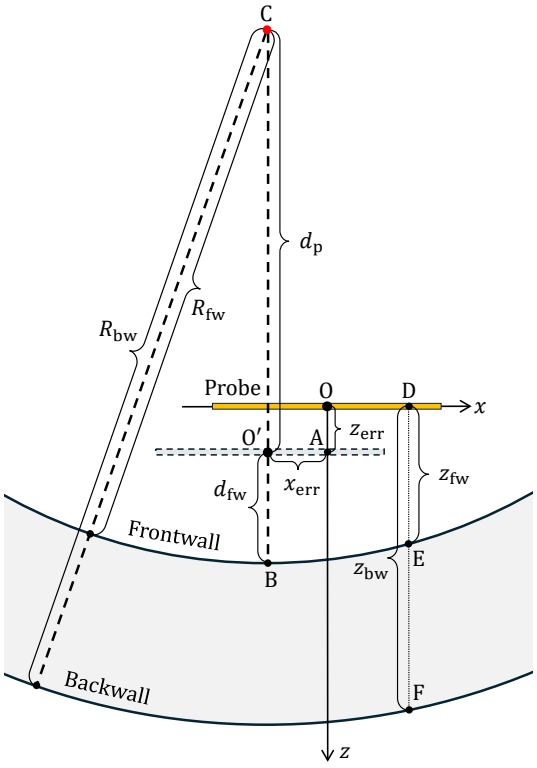


Fig. 1. Basic data acquisition geometry (not to scale) [8].

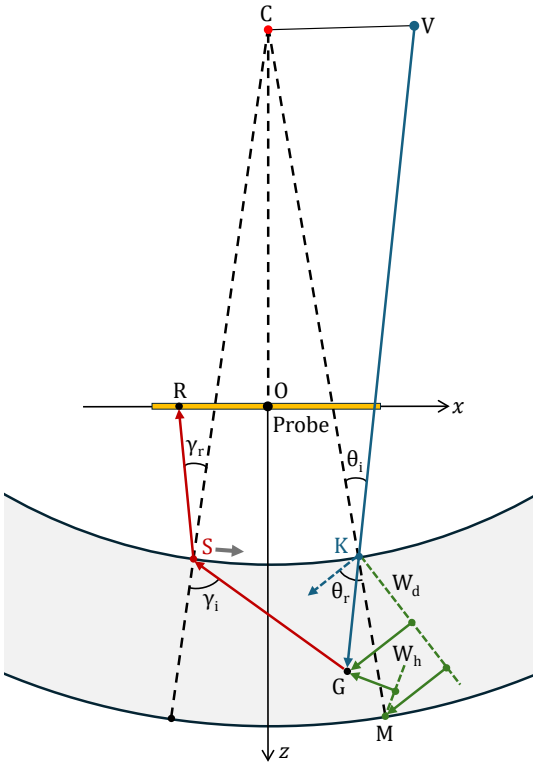


Fig. 2. Simplified virtual source model (negative DW angle).

4. Calculate approximate $\tau_{\text{TX}}^{\text{mode}}$ and $\tau_{\text{RX}}^{\text{mode}}$ for all the imaging modes of interest (i.e., L-L, T-T, and TT-T in this work).
5. Given acquired channel data $\mathcal{A}_\theta(t, x_e)$, reconstruct multi-mode views $\mathcal{I}_\theta(z, x)$ using (1) and (2).

The details for Steps 1–3 can be found in [8]. Our approach to accomplishing Step 4 is presented in sections II-A and II-B.

During a DW emission at a steering angle θ, the firing delays of individual array elements are such that the incidence angle with respect to the pipe frontwall curvature (i.e., at the liquid-steel interface) is nominally constant and equal to θ. Also, in our case:

- The firing delays (shaping emitted wavefronts) are based on the assumption of nominal operating conditions: $z_{\text{err}} = x_{\text{err}} = 0$ mm and $c_l = c_0$ (nominal SoS in liquid).
- When $\theta = 0°$, the diverging wavefront shape is circular, nominally matching the frontwall curvature upon impact (at zero-angle incidence).
- When $\theta < 0°$ or $\theta > 0°$, the diverging wavefront shape is not exactly circular to ensure constancy of the nominal nonzero incidence angle across the liquid-steel interface.
- We compensate for deviations from the nominal operating conditions only during image reconstruction.

Our virtual source model for DW emissions is illustrated in Fig. 2. It shows an example of the virtual source V for $\theta < 0°$; for $\theta > 0°$, point V would be located on the left side of the pipe center C. This is an approximation, because the wavefront generated by V would be circular, making the incidence angle θ vary across the frontwall boundary.[2] However, the resulting reconstruction errors are relatively small in the case of large-diameter pipes (see Section III). Next, we describe how $\tau_{\text{TX}}^{\text{mode}}$ and $\tau_{\text{RX}}^{\text{mode}}$ are calculated.

A. Calculation of Approximate Transmit-Path Delays

Let $\alpha(x) = \arcsin(x/R_{\text{fw}})$, which is the angle between the z-axis and radial line crossing a frontwall point with a given x coordinate. In Fig. 2, examples of α are $\angle$OCK (positive) and $\angle$OCS (negative). In this work, we limit the range of α to the interval $[\arcsin(x^{\min}/R_{\text{fw}}), \arcsin(x^{\max}/R_{\text{fw}})]$, corresponding to $[x^{\min}, x^{\max}] = [-L/2, L/2]$, where L is the lateral width of the array probe.

For the frontwall point K, the precise virtual source location is such that $\angle$VKC $= \theta$ and $|$VK$| = |$CK$|$, which implies that the placement of V depends on the x coordinate of K. To simplify our proposed model, we place a single virtual source at a fixed θ-specific location given by

$$(\Delta\bar{z}_\theta, \Delta\bar{x}_\theta) = \big(\text{mean}_\alpha[\Delta z_\theta(\alpha)], \text{mean}_\alpha[\Delta x_\theta(\alpha)]\big), \quad (3)$$

$$\Delta z_\theta(\alpha) = R_{\text{fw}}[\cos(\alpha) - \cos(\alpha + \theta)], \quad (4)$$

$$\Delta x_\theta(\alpha) = R_{\text{fw}}[\sin(\alpha) - \sin(\alpha + \theta)], \quad (5)$$

where $\Delta\bar{z}_\theta$ and $\Delta\bar{x}_\theta$ represent $|$VC$|$ projections onto the z-axis and x-axis, averaged over the range of α. In other words, the

[2]For $\theta = 0°$, the virtual source model is exact (with V placed at C in Fig. 2). Regardless of the model assumptions, the incidence angle could also vary across the frontwall boundary due to potential probe misalignment.

location of our now-fixed virtual source V becomes such that $\Delta\bar{z}_\theta^2 + \Delta\bar{x}_\theta^2 = |\mathtt{VC}|^2$.

Next, we consider an example target point G in Fig. 2. The transmit path length from V to G is modeled as $|\mathtt{VK}| + |W_d\mathtt{G}|$ (direct mode) or $|\mathtt{VK}| + |W_d\mathtt{M}| + |W_h\mathtt{G}|$ (half-skip mode). Symbol W_d refers to the plane wavefront created after refraction at the frontwall boundary, while symbol W_h refers to the plane wavefront after subsequent reflection at the backwall boundary. Hence, during reconstruction of the L-L, T-T, or TT-T views, our transmit-path delays for the target $\mathtt{G} = (z_\mathtt{G}, x_\mathtt{G})$ are

$$\tau_{\text{TX}}^{\text{L-}}(z_\mathtt{G}, x_\mathtt{G}, \theta) \approx |\mathtt{VK}|/c_l + |W_d\mathtt{G}|/c_{s\text{L}}, \tag{6}$$

$$\tau_{\text{TX}}^{\text{T-}}(z_\mathtt{G}, x_\mathtt{G}, \theta) \approx |\mathtt{VK}|/c_l + |W_d\mathtt{G}|/c_{s\text{T}}, \tag{7}$$

$$\tau_{\text{TX}}^{\text{TT-}}(z_\mathtt{G}, x_\mathtt{G}, \theta) \approx |\mathtt{VK}|/c_l + (|W_d\mathtt{M}| + |W_h\mathtt{G}|)/c_{s\text{T}}. \tag{8}$$

In Fig. 2, it can be shown that

$$|W_d\mathtt{G}| = (|\mathtt{VG}| - |\mathtt{VK}|)\cos(\theta_r - \theta_i), \tag{9}$$

$$|W_d\mathtt{M}| + |W_h\mathtt{G}| = 2(R_{\text{bw}} - R_{\text{fw}})\cos(\theta_r) - (|\mathtt{VG}| - |\mathtt{VK}|)\cos(\theta_r + \theta_i), \tag{10}$$

where $\theta_r = \arcsin((\{c_{s\text{L}}, c_{s\text{T}}\}/c_l)\sin(\theta_i))$ (Snell's law), and θ_i would ideally be the same as θ. However, due to potentially nonzero z_{err} and x_{err} (see Fig. 1) and fixed V, we have $\theta_i \approx \theta$, which is set to

$$\theta_i = \text{mean}_x[\arcsin((x - \Delta\bar{x}_\theta)/d_\mathtt{V}) - \arcsin((x + x_{\text{err}})/R_{\text{fw}})],$$

$$d_\mathtt{V} = \sqrt{(z_{\text{fw}} + d_p - \Delta\bar{z}_\theta)^2 + (x - \Delta\bar{x}_\theta)^2},$$

$$z_{\text{fw}} = \sqrt{R_{\text{fw}}^2 - (x + x_{\text{err}})^2} - d_{\text{p}} + z_{\text{err}}, \tag{11}$$

where z_{fw} denotes the z-coordinates of the frontwall boundary points, and $d_\mathtt{V}$ is the distance from V to those points. Finally, we adjust all $\tau_{\text{TX}}^{\text{mode}}$ by subtracting a fixed delay corresponding to the virtual travel distance from V to the array element that fires first.

B. Calculation of Approximate Receive-Path Delays

Unlike $\tau_{\text{TX}}^{\text{mode}}$, the receive-path delays are not affected by the proposed virtual source model. Instead, our approximations in this case stem from simplified handling of interface exit points S depicted in Fig. 2. Given the target point G, we sweep across the frontwall boundary points on the imaging grid and process them as follows. For each S, we determine the incidence and refraction angles γ_i and $\gamma_r = \arcsin((c_l/\{c_{s\text{L}}, c_{s\text{T}}\})\sin(\gamma_i))$, which identifies the return point R. The array element nearest to R is then assigned the corresponding receive-path delay from $\mathtt{G} = (z_\mathtt{G}, x_\mathtt{G})$ to $\mathtt{R} = (z_\mathtt{R}, x_\mathtt{R})$. Thus, during reconstruction of the L-L, T-T, or TT-T views, we use

$$\tau_{\text{RX}}^{\text{-L}}(z_\mathtt{G}, x_\mathtt{G}, x_e \approx x_\mathtt{R}) = |\mathtt{GS}|/c_{s\text{L}} + |\mathtt{SR}|/c_l, \tag{12}$$

$$\tau_{\text{RX}}^{\text{-T}}(z_\mathtt{G}, x_\mathtt{G}, x_e \approx x_\mathtt{R}) = |\mathtt{GS}|/c_{s\text{T}} + |\mathtt{SR}|/c_l. \tag{13}$$

To summarize, our approximations of $\tau_{\text{TX}}^{\text{mode}}$ and $\tau_{\text{RX}}^{\text{mode}}$ avoid searching for exact minimum-delay transmit and receive paths, which simplifies the DAS beamforming process.

III. Evaluation Results

To evaluate the accuracy of our method, we used three pipe samples depicted in Fig. 3 and labeled P_1, P_2, and P_3. Their backwall surface had crack-like defects of known dimensions. These samples were immersed in water and insonified with a 256-element probe (0.3-mm pitch) that emitted three 4.0-MHz DWs at $\theta = 0°, 20°, -20°$. The remaining parameters of our experimental setup are listed in Table I.

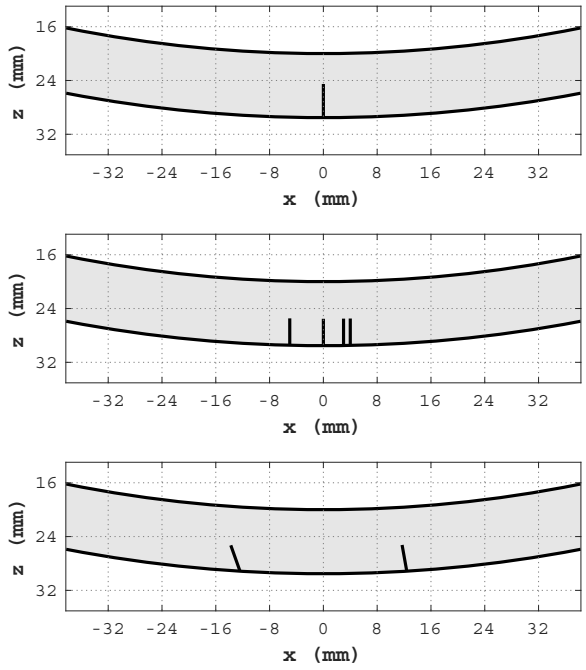


Fig. 3. Visualization of pipe samples P_1 (top), P_2 (middle), and P_3 (bottom).

TABLE I
Experimental Parameters

Parameter	Value
256-element probe width L	76.5 mm
DW emission angles θ	$0°, 20°, -20°$
DW central frequency f_c	4.0 MHz
Nominal standoff d_{fw}	20.0 mm
Pipe inner radius R_{fw}	193.7 mm
Pipe outer radius R_{bw}	203.2 mm
Nominal SoS in liquid c_l^{nom}	1480 m/s
Nominal transverse SoS in steel $c_{s\text{T}}^{\text{nom}}$	3120 m/s
Nominal longitudinal SoS in steel $c_{s\text{L}}^{\text{nom}}$	5640 m/s

During DAS beamforming, we set the pixel size to 0.1×0.1 mm^2. Fig. 4 shows the reconstructed multimode images, using the SoS values and probe positioning errors estimated based on [8] and listed below.

- Sample P_1: $(c_l, c_{s\text{L}}, c_{s\text{T}}) = (1470, 5710, 3130)$ m/s and $(z_{\text{err}}, x_{\text{err}}) = (2.0, -0.2)$ mm.
- Sample P_2: $(c_l, c_{s\text{L}}, c_{s\text{T}}) = (1460, 5640, 3080)$ m/s and $(z_{\text{err}}, x_{\text{err}}) = (-0.2, -0.8)$ mm.
- Sample P_3: $(c_l, c_{s\text{L}}, c_{s\text{T}}) = (1470, 5640, 3060)$ m/s and $(z_{\text{err}}, x_{\text{err}}) = (0.0, -2.1)$ mm.

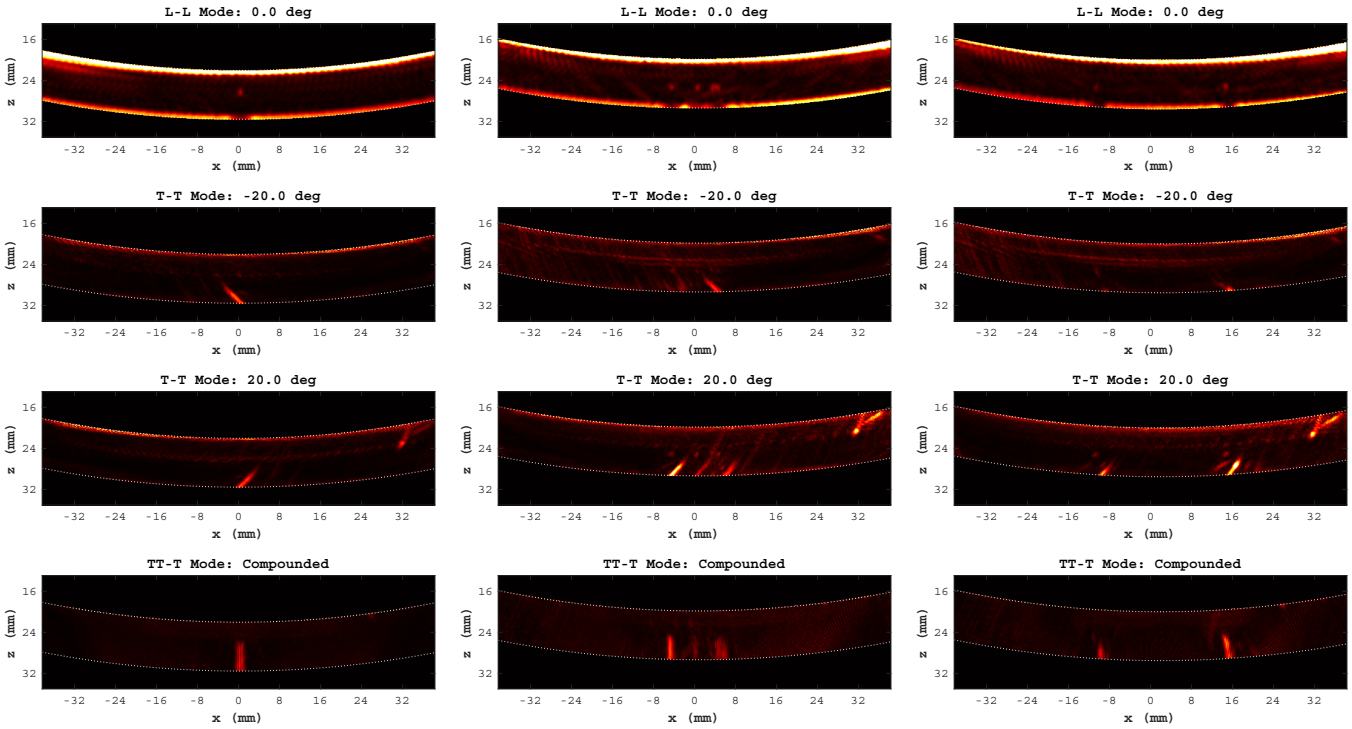


Fig. 4. Examples of reconstructed image envelopes of pipe samples P_1 (left), P_2 (center), and P_3 (right).

After DAS beamforming, we used the techniques described in [10] to determine imaged defect locations L_c and sizes S_c. Across the three pipe samples shown in Fig. 3, there are seven backwall cracks in total, and the localization accuracy for six of them was within 1.0 mm of their ground truth (GT) values. The worst localization error $|L_c - \text{GT}|$ (under 1.5 mm) was observed for the rightmost crack in P_2. We should note that the 1-mm clearance between this defect and its nearest neighbor (see Fig. 3) makes its precise localization difficult [10].

The sizing errors $|S_c - \text{GT}|$ were within 0.1 mm for five out of seven cracks under consideration. The other two cases were the rightmost defect in P_2 ($S_c = 3.2$ vs. GT $= 4.0$ mm) and the leftmost one in P_3 ($S_c = 1.7$ vs. GT $= 3.8$ mm) [10]. In the first case, the accuracy loss was primarily due to the close proximity of another 4-mm crack, causing interference that degraded imaging fidelity. In the second case, the sizing error was primarily due to the crack tilt of 20°, which distorted its true size in the reconstructed images (see Fig. 4) [10].

IV. Conclusion

We presented an approximate DAS beamforming method that simplifies multimode image reconstruction during in-line inspection of liquid-filled steel pipes. The proposed technique employs a virtual source model for transmit delay calculations and an approximate treatment of receive-path delays, enabling efficient reconstruction of L-L, T-T, and TT-T views.

Evaluation on three large-diameter pipe samples, containing seven manufactured backwall crack-like defects, showed that six of them were localized within 1.0 mm of the ground truth, with five crack size estimates being accurate within 0.1 mm. In the remaining cases, increased errors were attributed to signal interference from a nearby defect and distortion caused by a crack's tilt. Our future work will focus on addressing these limitations.

References

[1] M. Tanter and M. Fink, "Ultrafast imaging in biomedical ultrasound," *IEEE Trans. Ultrason., Ferroelect., Freq. Contr.*, vol. 61, no. 1, pp. 102–119, Jan 2014.

[2] L. Le Jeune, S. Robert, E. Lopez Villaverde, and C. Prada, "Plane wave imaging for ultrasonic non-destructive testing: Generalization to multimodal imaging," *Ultrasonics*, vol. 64, pp. 128–138, Jan 2016.

[3] R. Rachev, P. Wilcox, A. Velichko, and K. McAughey, "Plane wave imaging techniques for immersion testing of components with nonplanar surfaces," *IEEE Trans. Ultrason., Ferroelect., Freq. Contr.*, vol. 67, no. 7, pp. 1303–1316, Jul 2020.

[4] G. Cosarinsky, J. Fernandez-Cruza, and J. Camacho, "Plane wave imaging through interfaces," *Sensors*, vol. 21, p. 4967, Jul 2021.

[5] R. J. Pyle, R. R. Hughes, and P. D. Wilcox, "Interpretable and explainable machine learning for ultrasonic defect sizing," *IEEE Trans. Ultrason., Ferroelectr., Freq. Contr.*, vol. 70, no. 4, pp. 277–290, Apr 2023.

[6] L. Yu, X. Li, S. Yao, and H. Hu, "Ultrafast wide field-of-view ultrasonic diverging wave imaging based on delay p-th root and sum beamforming," *NDT & E Int.*, vol. 155, p. 103426, 2025.

[7] C. da Costa Filho, D. D'Souza, R. Zahiri, and G. Manders, "High-resolution multi-mode ultrasonic imaging of pipes using an ultrafast tool," in *Proc. IEEE Int. Ultrason. Symp. (IUS)*, Sep 2025, pp. 1–4.

[8] H. Chen, D. Rakhmatov, C. da Costa Filho, D. D'Souza, and R. Zahiri, "Sound speed and probe misalignment estimation for diverging-wave imaging of pipes," in *Proc. IEEE Int. Ultrason. Symp. (IUS)*, Sep 2025, pp. 1–5.

[9] V. Perrot, M. Polichetti, F. Varray, and D. Garcia, "So you think you can DAS? A viewpoint on delay-and-sum beamforming," *Ultrasonics*, vol. 111, p. 106309, Mar 2021.

[10] V. Govor, D. Rakhmatov, C. da Costa Filho, D. D'Souza, and R. Zahiri, "Localization and sizing of cracks in pipe walls using multimode ultrasound images," in *Proc. IEEE Int. Ultrason. Symp. (IUS)*, Sep 2025, pp. 1–5.